IBM blog. Suffice to say this model isn't just about predicting if it will rain tomorrow or not, it's much more than that. It's a flexible tool that can be fine-tuned for various applications, from local weather forecasting to global climate projections.

Apart from the core Prithvi WxC foundational AI model, IBM and NASA have also released two fine-tuned versions of it – for increasing the resolution of climate data for more accurate localised weather predictions, and another for studying how gravity waves affect Earth's atmosphere and cloud formation.

What's special about Prithvi WxC

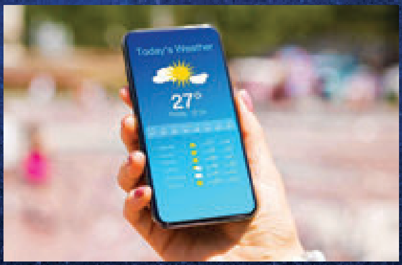
Traditional climate models are often large, resource-intensive, and focused on specific tasks. Prithvi WxC sets itself apart in several ways. Firstly, it's designed as a foundation model – which means it's a starting point that can be tweaked and adapted for multiple purposes. Similar to how large language models like GPT-3 function in natural language processing, for instance.

Another feather in its proverbial cap is that despite its complexity, the Prithvi WxC AI model is relatively small with just around 2.3 billion parameters. That's extremely light compared to some of the gargantuan AI models like Meta's Llama 3.1 or OpenAI's GPT-4 that can have hundreds of billion parameters. This efficiency makes the Prithvi AI model more accessible for organisations without supercomputing capabilities. And lastly, the weather and climate AI model is released on Hugging Face, a popular platform for sharing AI projects, thereby ensuring it is accessible to climate researchers and AI developers worldwide.

Needless to say, Prithvi WxC AI model ticks all the boxes for various people to try and build not just a weather app but even a climate-related AI chatbot – imagine how cool that would be!

Karen St. Germain, Director of NASA's Earth Science Division, highlights the importance of Prithvi's open-source accessibility. "The NASA foundation model will help us produce a tool that people can use – weather, seasonal, and climate projections to help inform decisions on how to prepare, respond, and mitigate," she emphasises.

Due to the foundational model nature of Prithvi WxC, there's several ways to deploy its game-changing AI sauce into derivative applications. By



fine-tuning the Prithvi climate model with local data, meteorologists at local weather forecasting stations can generate more accurate and targeted forecasts. Similarly, scientists and researchers can use the AI model for enhanced simulations of global climate conditions for better understanding long-term climate trends and, of course, climate change as well. In early experiments, the Prithvi WxC AI model accurately reconstructed global surface temperatures from just 5% of the original data, suggesting it can fill gaps where there isn't enough original data for weather estimates or climate projections.

Prithvi WxC's impact on climate science

Environment and Climate Change Canada (ECCC) is one of the early adopters of this new Prithvi climate AI model, whereby they're exploring its potential applications with the help of IBM. By ingesting real-time radar data, the model can generate very short-term precipitation forecasts. Additionally, it can improve forecast resolutions from 15 km down to the kilometre scale, enhancing weather prediction accuracy. Imagine something like this being deployed here in India? The value for accurate, hyper-local weather forecasts is immense in the fields of agriculture, disaster readiness, supply chain management, and logistics companies, just to name a few.

The open-sourcing of Prithvi WxC by NASA and IBM is more than a technical achievement – they're asking for the global community to act. Making advanced tools accessible ultimately empowers researchers, developers, and organisations to contribute to our collective understanding of the planet. In an era where climate change poses one of the most significant challenges to humanity, innovations like this offer a ray of hope. They enable us to predict, prepare for, and perhaps mitigate some of the impacts of a changing climate. d